

ECO 310 - Problem Set 2

Question 5(a) - Answer

Problem: Consider a decision-maker who expects to have a car accident with chance π ; if this occurs, he will incur $\$L$ in damages. He can purchase as much auto insurance, q , as he likes, at a price of p per dollar of coverage: this means that if he pays pq upfront (as the insurance premium), he'll receive a payment of q from the insurance company if an accident occurs.

Question: Write out his expected utility from purchasing insurance level q , assuming a utility-of-wealth function $u(w)$ and initial wealth w_0 .

Answer:

There are two possible states of the world:

1. **No accident** (probability $1 - \pi$):

- The decision-maker pays the insurance premium pq
- Final wealth: $w_0 - pq$
- Utility: $u(w_0 - pq)$

2. **Accident occurs** (probability π):

- The decision-maker pays the insurance premium pq
- Incurs damages of L
- Receives insurance payout of q
- Final wealth: $w_0 - pq - L + q = w_0 - L + q(1 - p)$
- Utility: $u(w_0 - L + q(1 - p))$

Therefore, the expected utility from purchasing insurance level q is:

$$\boxed{EU(q) = (1 - \pi)u(w_0 - pq) + \pi u(w_0 - L + q(1 - p))} \quad (1)$$

or equivalently:

$$\boxed{EU(q) = (1 - \pi)u(w_0 - pq) + \pi u(w_0 - L - pq + q)} \quad (2)$$